

Control Engineering: Maximizing Performance of the Fourth-Order Motion Setup

Sarthak Shirke 1658581, Jordy Schneiders 1633635 group 25

I. INTRODUCTION

For the project, an experimental setup was provided, for which the performance must be optimised. To do so, the concepts of control engineering need to be applied. This involves system identification, feedback controller design and implementation of feedforward. The motion setup consists of two rotating masses that are connected via a flexible axle. A motor is connected at the side of one of the masses, which will be called the collocated side. At both the collocated and the non-collocated side, an encoder is placed that senses the position of the mass. Firstly, the objectives of the project will be discussed. Afterwards, the experimental setup will be addressed briefly to get an understanding before considering the system identification. Subsequently, trajectory references for various periods were created complying with the given constraints. To get a stable system, a (basic) feedback controller is developed. That enabled tuning feedforward. Lastly, a performance controller is designed.

II. OBJECTIVE OF THE EXPERIMENT

This project aims to maximize the performance of the system while meeting the following requirements:

- The scanning stroke should be 120 rad (19.1 revolutions) both forward and backward
- The total allowable stroke is 125 rad, so there is only 2.5 rad of stroke for turn-around motion for both forward and backward motion.
- The modulus margin of the feedback loop should be at most 6 dB
- The sampling rate is restricted to at most 4 kHz

The objective of the experiment is to control the non-collocated mass with the following performance standards

Region	Period [sec]	RMS error [mrad]	peak error [mrad]
1	$10 < T_r \leq 12$	6	12
2	$10 < T_r \leq 11$	3	6.3
3	$10 < T_r$	3	6.3

TABLE I: Customer Requirements

For even further improved performance, it is desired to achieve a faster tracking period. This report will strive to maximize this performance.

III. EXPERIMENTAL SETUP

The research on 4-th-order motion setups has been quite of interest in the control field due to its wide implementations in control problems for companies. Applications that show very similar behaviour compared to the setup in this project are

wafer-steppers, (3D)-printers, pick-and-place machines and many more. The setup used in this project is referred to as the PATO setup in which there are 2 inertia connected via a torsion spring and one of the inertia is connected to an electric motor. A schematic of the setup is shown 1. Where J_1 and J_2 are the inertia, k is the stiffness of the torsion spring, d is a damping term, T_i is the input torque created by the motor and both the angles θ_1 and θ_2 can be measured via quadrature encoders which have a resolution of 0.003 [rad]. A DC motor is connected to one of the masses and two encoders are attached to each mass to measure revolutions. An electrical amplifier (E-Box) is connected to the experiment apparatus that constantly measures the input current supplied to the motor, this amplifier scales linearly with the torque that the motor generates. The system is linear with force as the input and the positioning of the masses as the output. The constraint of the system would be that the design of electrical circuitry and software implementations of communications between the computer and setup are all excluded from this project.

A. Dynamics of System

$$\begin{aligned} \ddot{x}_1 J_1 &= F - k(x_1 - x_2) - d(\dot{x}_1 - \dot{x}_2) \\ \ddot{x}_2 J_2 &= k(x_1 - x_2) + d(\dot{x}_1 - \dot{x}_2) \end{aligned} \quad (1)$$

in which x_1 and x_2 represent the positions of respectively the collocated and the non-collocated mass. F stands for the applied force, k for the spring constant and d for the damping constant.

From the expressions, the transfer functions can be derived. For the continuation, $H_{col}(s)$ is defined as the transfer function of the collocated mass, i.e. $x_1(s)$ over $F(s)$. $H_{ncol}(s)$ represents the transfer function of the non-collocated mass, which is $x_2(s)$ over $F(s)$.

$$\begin{aligned} H_{col}(s) &= \frac{1}{(J_1 + J_2)s^2} + \frac{\frac{J_1 J_2}{J_1 + J_2}}{J_1 J_2 s^2 + d(J_1 + J_2)s + k(J_1 + J_2)} \\ H_{ncol}(s) &= \frac{1}{(J_1 + J_2)s^2} - \frac{\frac{J_1 J_2}{J_1 + J_2}}{J_1 J_2 s^2 + d(J_1 + J_2)s + k(J_1 + J_2)} \end{aligned} \quad (2)$$

Equation 2 shows the formulas of the transfer functions. It can be observed that both contain similar terms. The

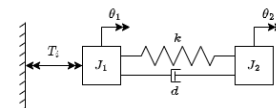


Fig. 1: System setup X

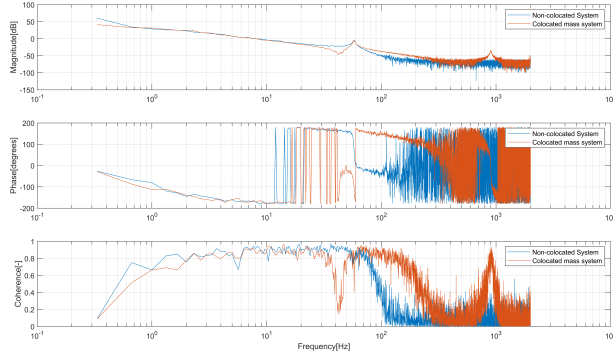


Fig. 2: FRF plot and comparison

first term is related to the so-called rigid-body mode, while the second term stems from the flexible mode. The corresponding bode plots will show a steeper slope for the non-collocated mass than for the collocated mass at sufficiently large frequencies. At low frequencies, however, the slope of the magnitude plot should be really comparable and have a value near -2. Also, because of the integrator term, a phase of 180 degrees is expected for low frequencies for both masses. These observations also must be visible in the measured FRF of the system.

IV. SYSTEM IDENTIFICATION

A. Decoding the Encoder: Unveiling the Inversion Process

As the plant dynamics were known beforehand from subsection III-A, the overall behaviour of the system could already be expected. Initially, the measurement were taken for the collocated mass that is attached to the motor. The observation is that it has an anti-resonance around 42 Hz and a resonance at 58 Hz. It can be seen that a -2 slope and a phase of -180 degrees at low frequencies could be noted as predicted from the transfer function. Subsequently, the non-collocated mass measurements were taken and it was observed that there is a resonance at a peak 58 Hz. At lower frequencies, in the magnitude plot, the expected -2 slope is present, however, the phase was at 0 degrees. This does not match with the bode gain-phase relation, as -2 to slope corresponds with a phase of -180 degrees. From this, it can be concluded that the encoder at the non-collocated side was inverted. Therefore it was decided that attaching a gain of -1 at the non-collocated mass side in Simulink was the most convenient. When multiplying a complex number with -1, the magnitude will remain the same while its phase will shift 180 degrees. Both the collocated and non-collocated plots are added in figure 2.

B. FRF Open loop Measurement

As for the observations done in subsection IV-A it is required to use direct method to get FRF. The FRF discussed previously was an initial measurement, meant to examine the reliability of the results. This paragraph focuses more on optimising the FRF measurement to determine the best

possible FRF for tuning controllers addressing the non-collocated mass FRF. The direct method was chosen for 2 reasons mainly controller was not required and the plant was stable. While doing the open loop identification it is important to cautiously select the parameters so that a better coherence is observed. The adjustable parameters are the number of samples, number of data frames, sample rate and noise input. Furthermore, filters can be applied to get smoother results. As a higher sample rate ensures a better result and as it is restricted to 4 kHz, this has been used for the continuation. For the noise input, a random normal distributed noise is taken, which allows for changing the mean and the variance. The mean is kept constant while the variance is varied. A larger variance means a larger excitation of the system, so it is recommended not to use too high values (not higher than 0.5), therefore was decided first to start with N-samples of 40,000 and a variance of 0.01. The FRF was noisy as the power (variance) was rather low. Next, the power was increased to 0.2 but still, the coherence was noisy. To obtain more distinguished plots, the number of samples is increased to 100,000 which would lead to better frames. The table provides the 100,000 samples parameter distribution II.

N_samples	Fs	Measurement time (seconds)	nfft (frame length in samples)	Resolution (Hz) = fs/nfft	Frame +nover-lap
100,000	4000	25 s	1,00,000	0.04	1
100,000	4000	25 s	2000	2	99
100,000	4000	25 s	12000	0.33	16

TABLE II: FRF Parameter Distribution

The observation with increase in N-samples to 100000 was that as the number of frames increases the loss of data at frame boundaries decreases. So for N-samples 100000 with noise 0.2, it had a better coherence than 40000 samples. To get a better FRF the variance was changed to 0.3 the FRF was better than the 0.2 variance. After this, it was decided to make the variance 0.5 but as per the indication, it was said to try less than 0.5 therefore a low-pass filter on the noise signal was considered at 100 Hz. It would reduce the magnitude of the input at high frequencies that are less relevant as the input designed for the experiment according to the requirement of the customer is nearly 3-6 Hz so the data above 100 Hz is not that relevant. This resulted in better FRF and coherence was nearly 1. But when this data was uploaded into shapeit it was observed that there were a lot of loops around the modulus margin at the higher frequency range due to the low-pass filter. So the actual plant and the model did not match well. Therefore the idea of adding a low-pass filter was dropped and it is recommended not to use one as a 'natural' identification of the system is desired.

Furthermore, the N-samples can be increased more to get a better overlapping which ultimately can decrease the data loss between the frames. But this increases the measurement time as well. Therefore the data was taken for 1,60,000 with a measurement time of 40 seconds and then finally for N-samples 2,00,000 with a measurement time of 50 seconds.

But the question was is it needed to go higher if the dynamics were visible with 100000 samples. Therefore it was decided to design a controller with N-samples of 100000 keeping the sample rate of 4000, measurement of 25 seconds. At start nfft was chosen 100000 which just gives 1 frame so the coherence is 1 there was no averaging and the noise is seen as the linear dynamics, therefore nfft was decreased to 2000 which gives the resolution of 2 with 99 frames but the resonance was damped and was not visible. Lastly, an nfft of 12,000 was chosen so the frames were 16 and the resolution was 0.33 Hz. This gave a good FRF with resonance not damped. In the end, the FRF was chosen to be used with the nfft of 12000 for control design and the comparison is shown in figure 3.

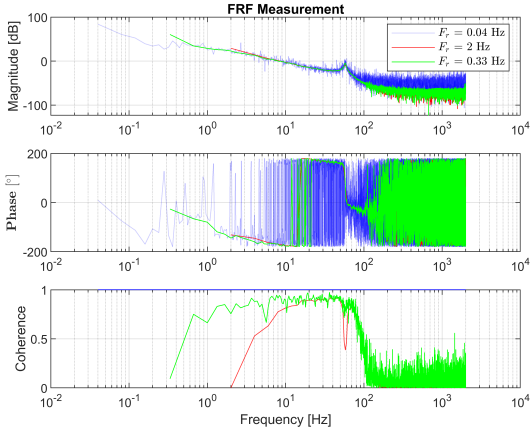


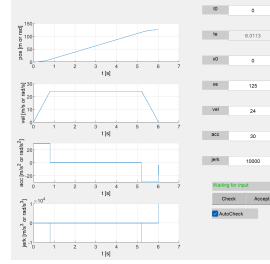
Fig. 3: Non-located FRF plot

V. SET POINT TRAJECTORY

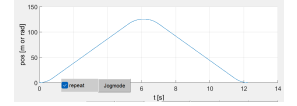
As we have the FRF the other requirement is trajectory to test the controller it is necessary to design the reference profile according to the requirements of the customer stated in section II.

A. Minimal Performance Trajectory 12 seconds

The 12s reference trajectory is designed for the minimal performance region which is between 10 to 12 seconds. For this, the ref generator in the Simulink was used. So for a 12-second trajectory, the total allowable stroke is 125 rad which is equivalent to 7161.97 degrees so the number of resolutions will turn out to be 19.89 approx. 20 revolutions. There should be a constant velocity of 120 rad that is 6875.49 degrees so it will be 19.0986 approx 19.01 resolutions. To have the constant velocity in the region of 120 rad the velocity is kept to 24 rad/s so per second we have 3.81 revolutions for nearly 5 seconds giving total 19 revolutions. To achieve this velocity the jerk was set to 10000 and acceleration to 30 as illustrated in figure 4. Similarly, the trajectories are designed for Improved tracking and Faster performance region.



(a) Forward motion of 12 seconds reference trajectory



(b) 12 seconds reference trajectory

Fig. 4: Profile 12 seconds

VI. CONTROLLER DESIGN

A. Closed loop stability

The plant on its own does not provide adequate stability margins to ensure stable adherence to the target provided by the customer as can be seen in Nyquist plot 10 (blue plot). Plant does not meet the margins but it is stable as there are no encirclements of the -1 point nor are there open-loop unstable poles, meaning that the closed-loop system is stable. Therefore, controllers are essential to enhance the bandwidth and reduce the margin of error. Such errors may arise from unexpected disturbances or shifts in the system's state of motion. In this project, the aim is to achieve precise positioning of the non-located mass along a two-way path. Thus, attention must be paid to both potential sources of error. As a result, a tuning methodology that incorporates both feed-forward and feedback control is employed.

Opting to control the non-co-located mass via the encoder on the load side is preferred to using the motor side encoder, as the latter is affected by anti-resonance that limits bandwidth. This limitation could be addressed by implementing an inverse notch filter; however, that presents a more complex solution.

B. Basic controller design

The Nyquist plot 10 (blue) shows the plant as it is observed where the plant is not able to meet any margins and the bandwidth is near 8.09 Hz as seen in 5. So a stabilizing



Fig. 5: Plant performance parameters

controller must be designed to achieve a modulus margin of less than 6db. Firstly a lead-lag was used, as it is observed from the 10 (blue) phase lead is needed to leave the -1 point on the left. Therefore it was decided to add a lead/lag controller and change the zero 8/2 and pole 8/2 which led to a phase lead but then by repeating and tuning the bandwidth/4, bandwidth/5 for zero and bandwidth*4, bandwidth*5 for pole the bandwidth was achieved to 23.2 Hz and the modulus margin were not meet so a notch filter was placed at zeros 58 with damping of 0.0195 and poles at 150 with

damping 1. This ratio of β_1/β_2 led to a decrease in the resonance (disturbance seen in the process sensitivity). The pole was placed far away from the resonance so that the zero could cancel the resonance. After which the big loop of the Nyquist plot was damped and with the small tuning of lead-lag that is zero at 6.6467 Hz and pole at 59.79 Hz made meet the margins. Then the Nyquist plot was not exactly leaving the -1 point but was entering the 4th quadrant. For this, it was observed that this might be because of the mass-line being low and that can be observed from the coherence plot 3 (green) that there is low coherence at the start so an integrator was added at 0.958 Hz such that the Nyquist plot entered leaving the -1 point on the left. The final performance parameters are shown in 6 and the Nyquist plot of a basic controller is seen in 10 (green).



Fig. 6: Basic controller performance parameters

In 6 it can be observed that the phase margin is shown as -321.5 degrees but looking at the open loop 10a (green) it is greater than 30 degrees. The basic controller filters can be found in III.

TABLE III

No.	Filters	Values
1	Gain	1.1
2	Lead/Lag	zero: 6.6467 Hz, pole: 59.79 Hz
3	Notch	zero and damp: 58 Hz & 0.0195, pole & damp: 150 Hz & 1
4	Integrator 2	zero: 0.958

VII. IMPLEMENTATION OF FEEDFORWARD

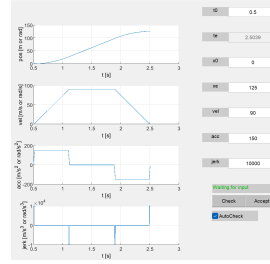
In control engineering, feed-forward is a proactive strategy to deal with expected disturbances by adjusting the system's output in advance based on anticipated changes. Unlike a feedback controller, which corrects deviations after they occur by monitoring the system's output, feed-forward control acts preemptively. It uses a known reference to anticipate errors and adjusts the system inputs accordingly, reducing the effects of lag that can occur with feedback mechanisms. For optimal performance, the feed-forward element should be designed as the precise opposite of the plant's dynamics.

A. Feedforward Profile design

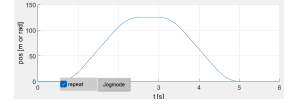
The Feedforward profile was designed to accurately see the error profile. So it was designed for 5 seconds with the approx 1/3 acceleration, 1/3 constant velocity and 1/3 deceleration as shown in 7. Keeping high acceleration and velocity in short amount of time.

B. Simple controller design for feedforward Tuning

To optimally tune feedforward, a simple feedback controller, that still ensures a stable system, is used. For this



(a) Forward motion of 5 seconds feedforward reference trajectory



(b) 5 seconds feedforward reference trajectory

Fig. 7: Profile feedforward 5 seconds

purpose, from the basic controller design, an integrator was removed as the acceleration (K_{fa}) term needs to be tuned. With the removed integrator the feedforward was tuned again and reached a bandwidth of 17.56 Hz. This controller was used to track the error but the error profile was not having a clear structure. Therefore the gain was reduced from 0.5 to 0.1 to get a bandwidth of 4.48 Hz. This showed the error profile clearly as shown in 8 (Untuned Error).

C. Feedforward Tuning

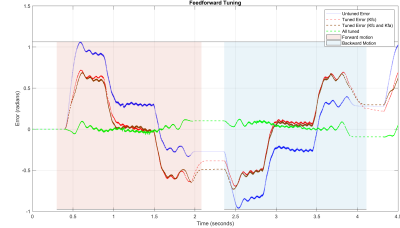


Fig. 8: Feed Forward Tuning

The first step is to tune the K_{fc} so the untuned error profile shown in the figure 8 where the deceleration movement and the acceleration movement were not the same which arises from ball bearings. Therefore a $K_{fc} = 0.031$ was applied to make the deceleration forward and acceleration backward movement have the same error. As there were no slopes in the region of constant velocity K_{fv} but to counteract linear friction, contributing to the overall controller and as the forward and backward constant velocity phases were not on the zero line K_{fv} was tuned with a value of 0.00001. After this tuning the only error observed is due to an acceleration, it is subsequently adjusted by K_{fa} to compensate for inertia during the desired acceleration so K_{fa} was tuned to 0.0004. This made a major improvement in the error profile. Finally, the feed-forward after tuning the parameters is $K_{fc} = 0.031$, $K_{fa} = 0.0004$ and $K_{fv} = 0.00001$. The aforementioned techniques yield the results depicted in Figure 8 and documented in the table IV.

VIII. DESIGN OF PERFORMANCE CONTROLLER

After the feedforward tuning VII-C and adding the basic controller design VI-B, the error reached 0.0135 rad RMS

TABLE IV: Feedforward tuning

k_{fc}	0.031
k_{fv}	0.0004
k_{fa}	0.00001

in the constant velocity region with a peak of 0.0377 rad. So to minimize the error further a PSD and CPSD of error was calculated to understand which frequency components present in the error had the maximum influence. The PSD and CPSD is shown in 9 (red graph).

But still not reached the customer requirement for that perfect tracking can be achieved when the attenuation is known. Then the relevant transfer function should counteract it. As per the PSD plot 9 (red graph), it can be understood that for 12 seconds trajectory there is a peak at 4 Hz so the internal model principle is used to get perfect tracking. For this with the help of an inverse notch filter with zero at 3 Hz with a damping of 0.5 and pole at 4 Hz with a damping of 0.05 which creates a resonance in the open loop 10a (red) and a dip in the sensitivity function as seen in the figure 10b (red). After this a low pass filter is added to cut off high frequent controller gain which is added at 200 Hz.

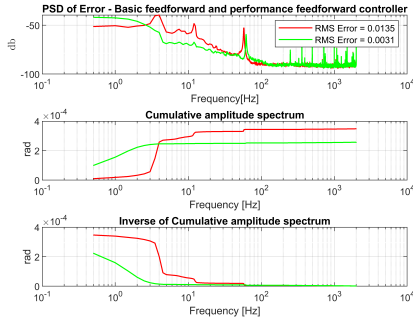
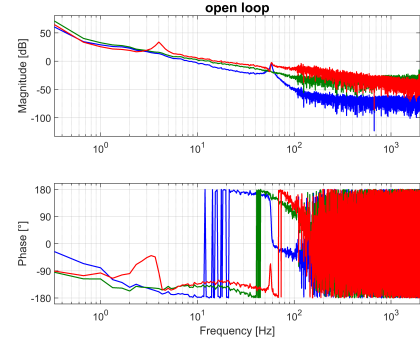


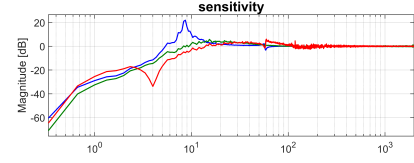
Fig. 9: PSD comparison basic feedforward to Performance feedforward

The process sensitivity (PS) = $\frac{H}{1+HC}d$, was observed to see where the incoming disturbance acts. The resonance is acting like a disturbance of a system. To damp it a narrow notch was used but it was seen that if a narrow notch was used then the controller would create disturbances which were observed in PSD and need to be lowered by using filters. Therefore if a wider notch was used then the controller would not create extra disturbance apart from the plant disturbance which has a resonance at 58 Hz. Therefore wider notch was used as shown in table V. This attenuation of a Notch filter creates a dip in the open loop 10a (red) and a dip can be observed in the closed loop (T) 10c (red). After this, the error was reduced and region 1 was achieved as shown in Table V and the summary is presented in VI. The controller designed for region 1 in the shapeit is compared to the discrete controller which runs on the raspberry pi and is shown in figure 11 (green) and it's not perfect at peaks but matches well overall.

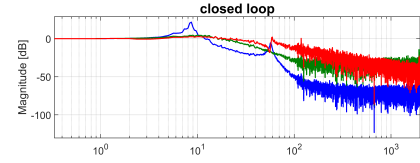
Finally, a PSD and CPSD was taken to see whether the



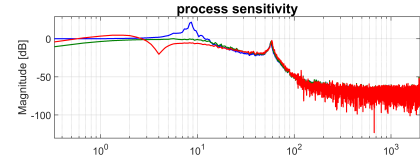
(a) Performance controller open loop



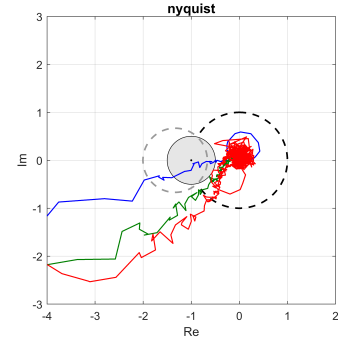
(b) Performance controller sensitivity



(c) Performance controller closed loop



(d) Performance controller Process sensitivity



(e) Performance controller Nyquist

Fig. 10: Reference and disturbance attenuation (Blue: Plant, Green: Basic Controller, Red: Performance Controller)

error has decreased and it can be observed from figure 9 (green) that the error has decreased in the region particularly at 4 Hz where the reference has a higher impact and near the disturbance at 58 Hz. After this the error dropped down to 3.1 mrad and the peak was 12.06 mrad achieving region 1. 12 second profile reference and the Comparison of the basic feedforward (ff), performance, and performance ff error is presented in the figure 12.

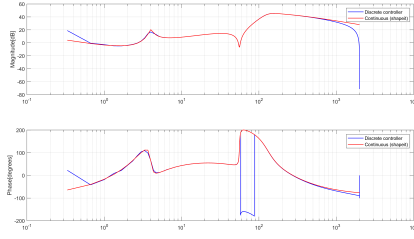


Fig. 11: Continuous and Discrete Controller Comparison

TABLE V: Performance controller Filters

No.	Filters	Values
1	Gain	0.348
2	Lead/Lag	zero: 2.6 Hz, pole: 200 Hz
3	Notch	zero and damping: 56 Hz & 0.016, pole & damp: 130 Hz & 0.3
4	Low pass 1st order	pole: 200 Hz
5	Notch	zero and damping: 3 Hz & 0.5, pole & damp: 4 Hz & 0.05
6	Integrator 2	zero: 1.5

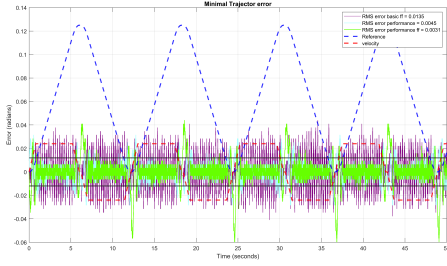


Fig. 12: Error profile comparison

The performance controller parameters are as follows and are meet as shown in 13.

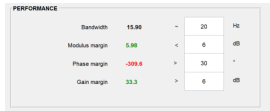


Fig. 13: Performance Controller Performance Parameters

IX. EXPERIMENTAL RESULTS

TABLE VI: Controller summary (Error constant velocity)

Controller	Error (rad)	Peak error
Basic (12-s)	0.0140	0.0346
Basic Feed Forward (12-s)	0.0135	0.0377
Performance (12-s)	0.0045	0.0126
Performance Feed Forward (12-s Region 1)	0.0031	0.0126
Performance Feed Forward (10.5-s (Close to Region 2))	0.0035	0.0094
Performance Feed Forward (9-s waveform (Far away to Region 3))	0.0044	0.0188

For a 12 s reference trajectory, the measured root-mean-squared error is 3.1 mrad and the peak error is 12.06 mrad.

This means the performances required for Region I are met. Subsequently, the results for a 10.5 s reference trajectory were observed. These were 3.5 mrad for the RMS error and 9.4 mrad for the peak error. Unfortunately, it does not meet the specifications necessary to be in Region II, however, the RMS error is not extremely far off. For completeness, the tracking error for a 9 s trajectory was considered as well. The tracking error was higher, with an RMS error of 4.4 mrad and a peak error of 18 mrad.

Previously, the focus was on the non-collocated mass. Also, the controller was made for that mass. However, as the collocated mass is easier to control, a controller was designed for collocated side as well. It was expected that by measuring the non-collocated mass, a better performance would be observed. Nevertheless, similar observations of the tracking error were made.

X. CONCLUSIONS & DISCUSSION

Conclusion

The dynamics of a fourth-order PATO system were modeled in the Laplace domain, focusing on phase and slope characteristics. The system's frequency response was analyzed using a direct method on both encoder sides, exploring various noise variances and nfft settings to determine the most reliable frequency range. A basic controller was initially designed, followed by adjustments in gain and the removal of an integrator to fine-tune feed-forward parameters. Despite viscous friction having minimal impact on error dynamics, the tuning of K_{fa} and K_{fc} gains was crucial. Subsequent error analysis through PSD and CPSD plots facilitated the design of a performance controller for optimal tracking. Ultimately, the integration of feed-forward and feedback control strategies significantly reduced the RMS error to 3.1 mrad and peak error to 12.06 mrad reaching first region.

Discussion

- The objectives for minimal performance in region 1 were met, achieving a peak error of 12.06 mrad and an RMS error of 3.1 mrad at a 4 kHz sample rate. Lowering the sample rate to 3 kHz reduced the peak error but then region 1 was not meet.
- The resolution of the encoder limits performance, it is not feasible to compensate for accuracies beyond the encoder's sensing capability.
- Increasing the number of samples beyond 100,000 could reduce data loss at frame boundaries by allowing for more overlaps. Future tests could compare the effectiveness of direct versus indirect methods for better FRF.
- Although the same experimental setup was used throughout, modifications between sessions could affect the FRF, though current results do not indicate any such alterations.

REFERENCES

- G.Witvoet, 4CM00, Control Engineering, (2023-2024)
- Wind Michiel. (n.d.). Motion Control on a fourth order system. Michiel Wind